

ASSIGNMENT-4.

Q1)

the kernel function is

$$k(x, x') = (1 + x^T x')^2$$

To find the corresponding transformation function

$$z = \phi(x)$$

As we know this kernel function comes under the polynomial example

Polynomial of power p : $k(x_i, x_j) = (1 + x_i^T x_j)^p$

For mapping $\phi: x \rightarrow \phi(x)$, where $\phi(x)$ has

$$\left(\frac{d+p}{p}\right) \text{ dimensions}$$

The function $\phi: x \rightarrow z$ is a polynomial of order p

kernel function: $k(x_i, x_j) = (1 + x_i^T x_j)^p$

The dimension of z is $\left(\frac{d+p}{p}\right)$

$$\phi(x) = [1, x_1^2, \sqrt{2}x_1x_2, x_2^2, \sqrt{2}x_1x_3, \sqrt{2}x_2x_3]^T$$

$$\Rightarrow [1, \sqrt{2}x_1, \sqrt{2}x_2, \sqrt{2}x_3, x_1^2, x_2^2, x_3^2, \sqrt{2}x_1x_2, \sqrt{2}x_1x_3, \sqrt{2}x_2x_3]^T$$

This transformation ensures that the dot product in the transformed space equals the kernel value.

Q2) HMM Probability calculation

Given hidden states F, B, B, F and observation sequence H, H, T, T, we calculate the joint probability:

$$P(\text{HHTT} | \text{FBBF}) = P(F) \cdot P(H|F) \cdot P(B|F) \cdot P(H|B) \cdot P(B|B) \cdot P(T|B) \cdot P(F|B) \cdot P(T|F)$$

Therefore

$$P(HHTT|FBBF) = 0.5 \times 0.5 \times 0.4 \times 0.75 \times 0.6 \times 0.25 \times 0.4 \times 0.5$$
$$= 0.00225$$

Q3)

Part 1: Manual Mutual Information calculation

We use:

$$MI(X; Y) = \sum_{x \in X} \sum_{y \in Y} P(x, y) \cdot \log_2 \left(\frac{P(x, y)}{P(x)P(y)} \right)$$

We'll do this for age, step by step, similar can be done for Income and Gender

A. Calculate frequencies for Age vs Risk

Age	High	low	Total
<25	4	0	4
≥25	1	3	4
Total	5	3	8

$$P(<25, \text{high}) = \frac{4}{8}$$

$$P(\geq 25, \text{high}) = \frac{1}{8}$$

$$P(\geq 25, \text{low}) = \frac{3}{8}$$

Also

$$P(<25) = \frac{4}{8} = 0.5$$

$$P(\geq 25) = \frac{4}{8} = 0.5$$

$$P(\text{high}) = \frac{5}{8}$$

$$P(\text{low}) = \frac{3}{8}$$

B. Compute MI (Age ; Risk)

$$MI = \sum P(x,y) \cdot \log_2 \left(\frac{P(x,y)}{P(x)P(y)} \right)$$

1) ($<25, \text{high}$):

$$= \frac{4}{8} \cdot \log_2 \left(\frac{\frac{4}{8}}{0.5 \cdot \frac{5}{8}} \right) = 0.5 \cdot \log_2 \left(\frac{0.5}{0.3125} \right)$$

$$= 0.5 \cdot \log_2 (1.6)$$

$$= 0.5 \times 0.6781$$

$$= 0.33905$$

2) (≥ 25 , high):

$$= \frac{1}{8} \cdot \log_2 \left(\frac{\frac{1}{8}}{0.5 \cdot \frac{5}{8}} \right) = 0.5 \cdot \log_2 \left(\frac{0.5}{0.3125} \right)$$

$$= 0.125 \cdot \log_2 \left(\frac{0.125}{0.5 \times 0.625} \right)$$

$$= 0.125 \cdot \log_2 \left(\frac{0.125}{0.3125} \right)$$

$$= 0.125 \cdot \log_2 (0.4)$$

$$= 0.125 \cdot (-1.322)$$

$$= -0.16525$$

3) (≥ 25 , low):

$$= \frac{3}{8} \cdot \log_2 \left(\frac{\frac{3}{8}}{0.5 \cdot \frac{3}{8}} \right)$$

$$= 0.375 \cdot \log_2 \left(\frac{0.375}{0.5 \times 0.375} \right)$$

$$= 0.375 \cdot \log_2 (2)$$

$$= 0.375$$

$$\text{sum} = 0.339 - 0.165 + 0.375 = 0.549$$

$$\text{do, } MI(\text{Age; Risk}) \approx \boxed{0.549}$$

Now we calculate ~~more~~

Now we see mutual information for income

A. Frequency Table: Income vs Risk

Income	high	low	Total
<50K	3	0	3
>50K	2	3	5
Total	5	3	8

$$P(<50K, \text{high}) = 3/8$$

$$P(>50K, \text{high}) = 2/8$$

$$P(>50K, \text{low}) = 3/8$$

$$P(<50K) = 3/8$$

$$P(>50K) = 5/8$$

$$P(\text{high}) = 5/8$$

$$P(\text{low}) = 3/8$$

B. Computing MI (Income; Risk)

1. (<50K, high):

$$= \frac{3}{8} \cdot \log_2 \left(\frac{\frac{3}{8}}{\frac{3}{8} \cdot \frac{5}{8}} \right)$$

$$= 0.375 \cdot \log_2 \left(\frac{0.375}{0.234375} \right)$$

$$= 0.375 \cdot \log_2(1.6)$$

$$= 0.375 \times 0.6781$$

$$= 0.2542875$$

2. (>50K, high):

$$= \frac{2}{8} \cdot \log_2 \left(\frac{\frac{2}{8}}{\frac{5}{8} \cdot \frac{5}{8}} \right)$$

$$= 0.25 \cdot \log_2 \left(\frac{0.25}{0.390625} \right)$$

$$= 0.25 \cdot \log_2(0.64)$$

$$= 0.25 \times (-0.6439) = -0.160975$$

$$3) (>50K, low): \frac{3}{8} = (low, M) \frac{1}{4}$$

$$= \frac{3}{8} \cdot \log_2 \left(\frac{\frac{3}{8}}{\frac{5}{8} \cdot \frac{3}{8}} \right) \cdot (low, \frac{1}{4})$$

$$= 0.375 \cdot \log_2 \left(\frac{0.375 \cdot \frac{1}{4}}{0.234375} \right)$$

$$= 0.375 \cdot \log_2(1.6)$$

$$= 0.2542875$$

$$Sum = 0.2542875 + 0.160975 + 0.2542875$$

$$= 0.347$$

$$So ME (Income; Risk) \approx \underline{0.347}$$

Now we do mutual information for gender

A. Frequency table

	High	Low	Total
M	2	2	4
F	3	1	4

$$P(M, \text{high}) = 2/8 \quad P(M, \text{low}) = 2/8 \quad P(F, \text{high}) = 3/8$$

$$P(F, \text{low}) = 1/8$$

B. Computing $I(\text{Gender}; \text{Risk})$

$$= 0.25 \log_2 \left(\frac{0.25}{0.5 \times 0.625} \right) + 0.25 \log_2 \left(\frac{0.25}{0.5 \times 0.375} \right)$$

$$+ 0.375 \log_2 \left(\frac{0.375}{0.5 \times 0.625} \right) + 0.125 \log_2 \left(\frac{0.125}{0.5 \times 0.375} \right)$$

$$= -0.080 + 0.104 + 0.099 - 0.073$$

$$= 0.05$$

Now we manually calculate the Chi-square statistic for each feature from the observed and expected frequencies

We already did frequency tables that was observed for age, income and gender

1. Age vs Risk

Expected (E):

$$E_{ij} = \frac{\text{row total} \times \text{column total}}{N}$$

$$E(<25, \text{high}) = \frac{4 \times 5}{8} = 2.5$$

$$E(<25, \text{low}) = \frac{4 \times 3}{8} = 1.5$$

$$E(\geq 25, \text{high}) = 2.5$$

$$E(\geq 25, \text{low}) = 1.5$$

Chi-Square Calculation

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$= \frac{(4 - 2.5)^2}{2.5} + \frac{(0 - 1.5)^2}{1.5} + \frac{(1 - 2.5)^2}{2.5} + \frac{(3 - 1.5)^2}{1.5}$$

$$= 0.9 + 1.5 + 0.9 + 1.5$$

$$= \boxed{4.8}$$

$$\frac{(258.1 - 5)^2}{259.1} + \frac{(258.1 - 5)^2}{258.1} + \frac{(251.8 - 5)^2}{251.8}$$

$$\frac{(251.1 - 1)^2}{251.1}$$

$$10.0 + 2900.0 + 2900.0 + 200.0 =$$

$$\boxed{10.032}$$

2. Income vs Risk

Observed(O) already given table

Expected (E):

$$E(<50K, \text{high}) = \frac{3 \cdot 5}{8} = 1.875$$

$$E(<50K, \text{low}) = \frac{3 \cdot 3}{8} = 1.125$$

$$E(>50K, \text{high}) = 3.125$$

$$E(>50K, \text{low}) = 1.875$$

Chi Square:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$= \frac{(3-1.875)^2}{1.875} + \frac{(0-1.125)^2}{1.125} + \frac{(2-3.125)^2}{3.125} + \frac{(3-1.875)^2}{1.875}$$

$$= 0.675 + 1.125 + 0.405 + 0.675 = \boxed{2.88}$$

Chi-Square: Gender

$$E(M, high) = \frac{4 \cdot 5}{8} = 2.5$$

$$E(M, low) = \frac{4 \cdot 3}{8} = 1.5$$

$$E(F, high) = 2.5$$

$$E(F, low) = 1.5$$

Chi-Square

$$\chi^2 = \frac{(2-2.5)^2}{2.5} + \frac{(2-1.5)^2}{1.5} + \frac{(3-2.5)^2}{2.5} + \frac{(1-1.5)^2}{1.5}$$

$$= \frac{0.25}{2.5} + \frac{0.25}{1.5} + \frac{0.25}{2.5} + \frac{0.25}{1.5} = 0.5334$$

$$\left(\frac{0.152}{0.2 \times 0.352} \right) + \left(\frac{0.352}{0.2 \times 0.352} \right) + \left(\frac{0.152}{0.2 \times 0.352} \right) + \left(\frac{0.352}{0.2 \times 0.352} \right)$$

$$= 0.005 + 0.005 + 0.005 + 0.005 = 0.02$$